Flow to a well of finite diameter in a homogeneous, anisotropic water table aquifer

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Abstract. A Laplace transform solution is presented for the problem of flow to a partially penetrating well of finite diameter in a slightly compressible water table aquifer. The solution, which allows for evaluation of both pumped well and observation piezometer data, accounts for effects of well bore storage and skin and allows for the noninstantaneous release of water from the unsaturated zone. For instantaneous release of water from the unsaturated zone the solution approaches the line source solution derived by Neuman as the diameter of the pumped well approaches zero. Delayed piezometer response, which is significant during times of rapidly changing hydraulic head, is included in the theoretical treatment and shown to be an important factor in accurate evaluation of specific storage. By means of a hypothetical field example it is demonstrated that evaluations of specific storage (S_s) using classical line source solutions may yield values of S_s that are overestimated by a factor of 100 or more, depending upon the location of the observation piezometers and whether effects of delayed piezometer response are included in the analysis. Theoretical responses obtained with the proposed model are used to suggest methods for evaluating specific storage.

Introduction

Although flow to a well in a water table aquifer has been modeled extensively, it appears that none of the available analytical solutions take into consideration effects arising from the finite diameter of the pumped well. The classical solutions developed by *Boulton* [1954], *Dagan* [1967], and *Neuman* [1972, 1974] assume the pumped well to be infinitesimal in diameter. This apparent gap in the literature is surprising in view of the fact that existing solutions cannot be used to correctly interpret early time pumped well and piezometer data. Also, although the Boulton and Neuman solutions account for effects of aquifer specific storage, the parameter cannot be accurately evaluated without taking special precautions to eliminate effects of well bore storage from the test data. In fact, as demonstrated by *Mucha and Paulikova* [1986], effects of well bore storage in both the pumped well and the observation piezometers can easily account for the 1–2 orders of magnitude overestimation of specific storage sometimes found in the published literature.

Kipp [1973] developed a solution that accounts for the finite diameter of the pumped well but assumed the water to be incompressible and the porous matrix to be rigid. Consequently, although well bore storage is included, the aquifer specific storage is not. *Boulton and Streltsova* [1976] also developed a solution for a water table aquifer that takes well bore storage into account. Their solution allows for aquifer compressibility but, because they assume the water table to be a constant head boundary, they cannot properly account for effects of specific yield. Unfortunately, their solution is valid only for a short initial time period when effects of well bore storage dominate the response to pumping. Perhaps because of perceived difficulties in obtaining an analytical solution to this problem, several investigators have developed numerical models to account for effects of a finite diameter pumped well [e.g., *Sridharan et al.*, 1990; *Narasimhan and Zhu*, 1993]. *Narasimhan and Zhu* [1993] used their model to demonstrate, among other things, that effects of well bore storage in the pumped well can mask large parts of the early time and intermediate time responses seen in *Neuman*'s [1975] type curves.

It has been demonstrated in the literature [see e.g., *Moench*, 1994] that the *Neuman* [1972, 1974] model, properly applied, can be used to estimate the most important water table aquifer parameters (horizontal hydraulic conductivity, vertical hydraulic conductivity, and specific yield) with reasonable accuracy. As stated above, however, accurate estimates of specific storage are often not possible with the Neuman model because they require use of early time drawdown (or recovery) data.

It is the purpose of this paper to present an exact solution to the problem of flow to a partially penetrating well of finite diameter in a water table aquifer. The solution is obtained in Laplace space and inverted numerically using the *Stehfest* [1970] algorithm. Theoretical responses and a hypothetical example are used to demonstrate the influence of well bore storage and its effect upon the evaluation of specific storage. In addition, because of the importance of early time data in the evaluation of specific storage, the delayed response (time lag) of observation piezometers is included in the theoretical development. This is accomplished by using a modification of the approach of *Black and Kipp* [1977].

Mathematical Model

Assumptions

As with all mathematical models, several simplifying assumptions are required. Most of the assumptions are identical to those of *Neuman* [1974] and are listed here as follows: (1) The aquifer is homogeneous, infinite in lateral extent, horizontal, and of uniform thickness. (2) The aquifer can be anisotro-

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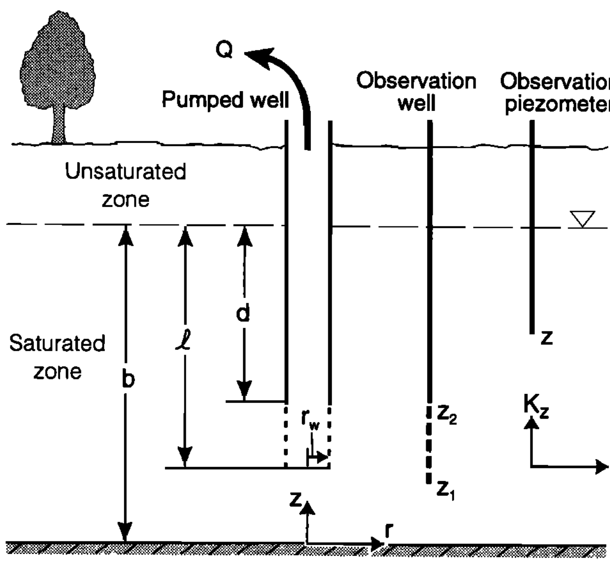

Figure 1. Schematic diagram of a finite diameter pumped well, observation well, and observation piezometer in a laterally extensive, homogeneous, anisotropic water table aquifer.

pic provided that the principal directions of the hydraulic conductivity tensor are parallel to the coordinate axes. (3) Vertical flow across the lower boundary of the aquifer is negligible (i.e., the lower boundary is impermeable). (4) A well discharges at a constant rate from a specified zone below an initially horizontal water table. (5) The change in saturated thickness of the aquifer due to pumping is small compared with the initial saturated thickness. (6) The porous medium and fluid are slightly compressible and have constant physical properties. (7) The initial hydraulic head is the same everywhere.

The assumptions pertaining to the finite-diameter pumped well are identical to those of *Dougherty and Babu* [1984] and are listed here as follows: (1) The head within the well does not vary spatially. (2) The radial flux from the aquifer to the well does not vary along the length of the screened section. A discussion of the approximation resulting from this assumption is given by *Hantush* [1964, pp. 351–352]. The physical inconsistency between these first two assumptions is deemed to be of negligible consequence in this and most well-hydraulics problems. (3) Vertical flux from the aquifer through the base of the well is negligible. (4) A thin skin of low-permeability material having no significant storage capacity may be present at the interface between the well screen and the aquifer. Well bore skin may be present for a number of reasons such as, for example, restricted flow due to the well screen itself, bridging by sand particles across screen openings, or damage to the aquifer caused by drilling.

Often overlooked in the analysis of pumping tests is the influence of a delayed response of the observation piezometers. The effect is treated approximately in this paper (following *Black and Kipp* [1977]) by assuming that the hydraulic head in an observation piezometer changes with a rate that is proportional to the head difference between the piezometer and the adjacent aquifer.

It is also assumed that the vertical flux of water into the aquifer from the unsaturated zone occurs in a manner that varies exponentially with time in response to a step decline in hydraulic head at the water table. The rate of exponential decline is controlled by a relaxation coefficient α_1 [see *Moench*, 1995]. (The subscript on α is included to avoid confusion with *Boulton*'s [1963] reciprocal of "delay index," which has a meaning that is slightly different from the α_1 used here. The difference is due to the fact that Boulton worked with vertically averaged heads and included the term containing α in the governing partial differential equation.) The assumption provides only a crude approximation of the actual drainage process [*Narasimhan and Zhu*, 1993], but it is a significant improvement over the assumption of instantaneous drainage (where α_1 is infinite) used by *Neuman* [1972, 1974].

Figure 1 is a diagrammatic cross section through a part of an idealized water table aquifer with a finite diameter partially penetrating pumped well, an observation well, and an observation piezometer. Figure 1 illustrates the parameters used to define the location of well screens, the location of observation points, and the anisotropic character of hydraulic conductivity. Also shown is the location of the origin of the coordinate system. Symbols used in the mathematical development are defined in the Notation section.

Boundary-Value Problem

The governing equation in the domain $r_w \leq r \leq \infty$ and $0 \leq z \leq b$ for axisymmetric flow to a pumped well in a slightly compressible, anisotropic, unconfined aquifer may be written as

$$\frac{\partial^2 h}{\partial r^2} + \frac{1}{r}\frac{\partial h}{\partial r} + \frac{K_z}{K_r}\frac{\partial^2 h}{\partial z^2} = \frac{S_s}{K_r}\frac{\partial h}{\partial t} \tag{1}$$

The initial condition in the domain of (1) is

$$h_i - h(r, z, 0) = 0 \tag{2}$$

where h_i is the initial hydraulic head. The outer boundary condition (at $r = \infty$) is

$$h_i - h(\infty, z, t) = 0 \tag{3}$$

The inner boundary condition requires a well bore balance equation for a partially penetrating well. Following *Dougherty and Babu* [1984], this condition is (at $r = r_w$)

$$2\pi r_w(l - d)K_r \left.\frac{\partial h}{\partial r}\right|_{r=r_w} = Q + C\frac{\partial h_w}{\partial t} \tag{4a}$$

$$b - l \leq z \leq b - d$$

where $l - d$ is the length of the screen, Q is the pumping rate, C is the well bore storage (assumed constant), and h_w is the average head in the well bore. The radial flow from the aquifer to the well ($K_r\partial h/\partial r$) at the sand face ($r = r_w$) is assumed to be independent of z and varies only with time. *Ramey and Agarwal* [1972] point out that effects of well bore storage can occur as a result of changing liquid level in the well or by virtue of well bore fluid compressibility in a pressurized test. Effects of well bore storage are greatest when resulting from changing water level. In this case, C is the cross-sectional area of the free surface in the well. In this paper, for convenience, $C = \pi r_c^2$, where r_c is the radius of the well in the interval where water levels are changing. A pressurized test might not be practical for a water table aquifer, but the influence of well bore storage could be reduced if a packer were placed slightly above the screen (so that the water table does not decline below the packer in the course of the test) and water level measurements were made with a transducer placed in or below a small diameter standpipe that penetrates the packer.

The spatial average of the head in the well bore, h_w, is related to the average head in the aquifer adjacent to the pumped-well screen, h^*, by

$$K_s \frac{(h^* - h_w)}{d_s} = K_r \left.\frac{\partial h}{\partial r}\right|_{r=r_w} \tag{4b}$$

where K_s is the hydraulic conductivity of the well bore skin, d_s is the skin thickness, and h^* is defined by

$$h^* = \frac{1}{(l-d)} \int_{b-l}^{b-d} h(r_w, z, t)\, dz \tag{4c}$$

Equation (4b) derives from the heat flow literature [*Carslaw and Jaeger*, 1959, p. 20]. The storage capacity of the skin is assumed to be negligible in this paper. The one additional equation required to complete the boundary condition at the interface between the pumped well and the aquifer is

$$\left.\frac{\partial h}{\partial r}\right|_{r=r_w} = 0 \qquad z < b - l;\ z > b - d \tag{4d}$$

Condition (4d) implies that a well casing of constant external radius r_w extends from the top of the screened section to the water table and from the bottom of the screened section to the base of the saturated thickness of the aquifer.

The condition along the base of the aquifer ($z = 0$) for $r \geq r_w$ is that of a no-flow boundary and is written as

$$\frac{\partial h}{\partial z}(r, 0, t) = 0 \tag{5}$$

and the condition at the water table ($z = b$) for $r \geq r_w$, which approximates the rate of drainage per unit area from the unsaturated zone, is written as

$$K_z \frac{\partial h}{\partial z}(r, b, t) = -\alpha_1 S_y \int_0^t \frac{\partial h}{\partial t'}(r, b, t) \exp\left[-\alpha_1(t - t')\right] dt' \tag{6}$$

Details pertaining to the theoretical justification for the use of (6), which derives from the work of *Boulton* [1954], are provided by *Moench* [1995]. If $\alpha_1 \rightarrow \infty$, (6) becomes the boundary condition used by *Neuman* [1972, 1974]. It can be seen by inspection that if $\alpha_1 = 0$, a no-flow condition is obtained for (6) and the boundary-value problem reduces to that solved by *Dougherty and Babu* [1984] for a confined, single-porosity aquifer.

By substituting the dimensionless parameters listed in Table 1 into (1)–(6), one obtains the following dimensionless boundary-value problem:

$$\frac{\partial^2 h_D}{\partial r_D^2} + \frac{1}{r_D}\frac{\partial h_D}{\partial r_D} + \beta_w \frac{\partial^2 h_D}{\partial z_D^2} = \frac{\partial h_D}{\partial t_D} \tag{7}$$

$$h_D(r_D, z_D, 0) = 0 \tag{8}$$

$$h_D(\infty, z_D, t_D) = 0 \tag{9}$$

$$-\frac{\partial h_D}{\partial r_D} = \frac{2}{(l_D - d_D)} - W_D \frac{\partial h_{wD}}{\partial t_D} \qquad 1 - l_D \leq z_D \leq 1 - d_D \quad r_D = 1 \tag{10a}$$

Table 1. Dimensionless Parameters and Expressions

Dimensionless Parameter	Expression
t_D	$K_r t/r_w^2 S_s$
t_{Dy}	$K_r b t/r_w^2 S_y$
h_D	$4\pi K_r b(h_i - h)/Q$
h_{mD}	$4\pi K_r b(h_i - h_m)/Q$
h_{wD}	$4\pi K_r b(h_i - h_w)/Q$
r_D	r/r_w
r_{wD}	r_w/b
z_D	z/b
K_D	K_z/K_r
l_D	l/b
d_D	d/b
β_w	$K_D r_{wD}^2$
β	$\beta_w r_D^2$
σ	$S_s b/S_y$
γ	$\alpha_1 b S_y/K_z$
W_D	$\pi r_c^2/[2\pi r_w^2 S_s(l - d)]$
W'_D	$\pi r_p^2/(2\pi r_w^2 S_s F')$
S_w	$K_r d_s/K_s r_w$

$$\frac{\partial h_D}{\partial r_D} = 0 \qquad z_D < 1 - l_D;\ z_D > 1 - d_D \quad r_D = 1 \tag{10b}$$

$$h_{wD} = h_D^* - S_w \frac{\partial h_D}{\partial r_D} \qquad 1 - l_D \leq z_D \leq 1 - d_D \tag{10c}$$

$$h_D^* = \frac{1}{(l_D - d_D)} \int_{1-l_D}^{1-d_D} h_D(1, z_D, t_D)\, dz_D \tag{10d}$$

$$\frac{\partial h_D}{\partial z_D}(r_D, 0, t_D) = 0 \tag{11}$$

$$\frac{\partial h_D}{\partial z_D}(r_D, 1, t_D) = -\gamma \int_0^{t_D} \frac{\partial h_D}{\partial \tau}(r_D, 1, t_D) e^{-\gamma\sigma\beta_w(t_D - \tau)}\, d\tau \tag{12}$$

Laplace Transform Solution

Application of the method of Laplace transformation to (7)–(12) leads to the following subsidiary boundary-value problem:

$$\frac{\partial^2 \bar{h}_D}{\partial r_D^2} + \frac{1}{r_D}\frac{\partial \bar{h}_D}{\partial r_D} + \beta_w \frac{\partial^2 \bar{h}_D}{\partial z_D^2} = p\bar{h}_D \tag{13}$$

$$\bar{h}_D(\infty, z_D, p) = 0 \tag{14}$$

$$\frac{\partial \bar{h}_D}{\partial r_D} = -\frac{2}{p(l_D - d_D)} + W_D p \bar{h}_{wD} \qquad 1 - l_D \leq z_D \leq 1 - d_D \quad r_D = 1 \tag{15a}$$

$$\frac{\partial \bar{h}_D}{\partial r_D} = 0 \qquad z_D < 1 - l_D;\ z_D > 1 - d_D \quad r_D = 1 \tag{15b}$$

$$\bar{h}_{wD} = \bar{h}_D^* - S_w \frac{\partial \bar{h}_D}{\partial r_D} \qquad 1 - l_D \leq z_D \leq 1 - d_D \tag{15c}$$

$$\bar{h}_D^* = \frac{1}{(l_D - d_D)} \int_{1-l_D}^{1-d_D} \bar{h}_D(1, z_D, p)\, dz_D \tag{15d}$$

$$\frac{\partial \bar{h}_D}{\partial z_D}(r_D, 0, p) = 0 \tag{16}$$

$$\frac{\partial \bar{h}_D}{\partial z_D}(r_D, 1, p) = -\frac{\bar{h}_D(r_D, 1, p)p}{\sigma\beta_w + p/\gamma} \tag{17}$$

When $\gamma = \infty$, (17) reduces to the subsidiary water table boundary condition for instantaneous release of water from the unsaturated zone, assumed by *Neuman* [1972, 1974].

The solution to the above subsidiary boundary-value problem is obtained with the help of Fourier cosine series in the manner followed by *Dougherty and Babu* [1984] for well tests in confined double-porosity aquifers. The Laplace transform solution (see Appendix for a derivation) for the dimensionless drawdown in the pumped well is

$$\bar{h}_{wD}(p) = \frac{2(A + S_w)}{p(l_D - d_D)[1 + W_D p(A + S_w)]} \tag{18}$$

and the solution for the dimensionless drawdown in the aquifer is

$$\bar{h}_D(r_D, z_D, p) = \frac{2E}{p(l_D - d_D)[1 + W_D p(A + S_w)]} \tag{19}$$

where

$$A = \frac{2}{(l_D - d_D)} \cdot \sum_{n=0}^{\infty} \frac{K_0(q_n)\{\sin[\varepsilon_n(1 - d_D)] - \sin[\varepsilon_n(1 - l_D)]\}^2}{\varepsilon_n q_n K_1(q_n)[\varepsilon_n + 0.5\sin(2\varepsilon_n)]} \tag{20}$$

$$E = 2\sum_{n=0}^{\infty} \{K_0(q_n r_D)\cos(\varepsilon_n z_D)[\sin(\varepsilon_n(1 - d_D)) - \sin(\varepsilon_n(1 - l_D))]\} \cdot \{q_n K_1(q_n)[\varepsilon_n + 0.5\sin(2\varepsilon_n)]\}^{-1} \tag{21}$$

$$q_n = (\varepsilon_n^2 \beta_w + p)^{1/2} \tag{22}$$

$$q_n r_D = (\varepsilon_n^2 \beta + p r_D^2)^{1/2} \tag{23}$$

and ε_n, where $n = 0, 1, 2, \cdots$, are the roots of

$$\varepsilon_n \tan(\varepsilon_n) = \frac{p}{(\sigma\beta_w + p/\gamma)} \tag{24}$$

For a long-screened observation well it is assumed that the measured drawdown is the average drawdown over the screened interval z_{D2}–z_{D1} as determined by

$$\bar{h}_D(r_D, z_{D1}, z_{D2}, p) = \frac{1}{z_{D2} - z_{D1}} \int_{z_{D1}}^{z_{D2}} \bar{h}_D(r_D, z_D, p)\, dz_D \tag{25}$$

Dimensionless drawdown then becomes

$$\bar{h}_D(r_D, z_{D1}, z_{D2}, p) = \frac{2E'}{p(l_D - d_D)[1 + W_D p(A + S_w)]} \tag{26}$$

where

$$E' = 2\sum_{n=0}^{\infty} \frac{K_0(q_n r_D)[\sin(\varepsilon_n z_{D2}) - \sin(\varepsilon_n z_{D1})]}{(z_{D2} - z_{D1})\varepsilon_n q_n K_1(q_n)[\varepsilon_n + 0.5\sin(2\varepsilon_n)]} \cdot \{\sin[\varepsilon_n(1 - d_D)] - \sin[\varepsilon_n(1 - l_D)]\} \tag{27}$$

Dimensionless drawdown is obtained by numerical inversion of (18) and (19) or (26) by means of the *Stehfest* [1970] algorithm in the same way as was accomplished by *Moench* [1996] for flow to a partially penetrating well of infinitesimal diameter in a water table aquifer. The FORTRAN code used for this paper (WTAQ3) is available from the author upon request.

Delayed Response of Observation Piezometers

Equations (19) and (26) are solutions for head changes in the aquifer. It is generally assumed that piezometers in good hydraulic connection with the aquifer respond to head changes instantaneously. In fact, however, it takes a finite time for a piezometer to dissipate stored water and to come to equilibrium with the hydraulic head in the aquifer [*Hvorslev*, 1951]. Ordinarily, delayed piezometer response is not a problem for evaluation of hydraulic conductivity and specific yield because estimates of these parameters do not depend upon the rapidly changing, early time drawdown data. Evaluation of specific storage, however, does depend upon early time drawdown data. Proper treatment requires not only inclusion of the effects of the finite diameter of the pumped well but also inclusion of delayed piezometer response. In this paper, delayed piezometer response is treated using the approach of *Black and Kipp* [1977].

In the notation of this paper the differential equation derived by *Hvorslev* [1951] can be written as

$$\pi r_p^2 \frac{dh_m}{dt} + F'K_r h_m = F'K_r h \tag{28}$$

where r_p is the radius of the piezometer standpipe in the interval over which water levels are changing, h_m is the measured hydraulic head, and $F' = F/2\pi$ where F is a shape factor defined by *Hvorslev* [1951]. The initial condition is $h = h_m = 0$. It is clear from (28) that if the aquifer is highly permeable and/or the radius r_p is small, the effects of delayed piezometer response might be negligible. Indeed, *Lissey* [1967] suggests the use of an insertible packer assembly to reduce standpipe diameter and thereby increase piezometer sensitivity. Similarly, *Butler et al.* [1996] used a transducer placed below a packer located just above the screen and effectively reduced the standpipe radius to zero. It should be noted that (28) is an approximate equation in that it neglects effects of aquifer compressibility and drawdown of the free surface (water table).

With the dimensionless parameters defined in Table 1, (28) becomes

$$W'_D \frac{dh_{mD}}{dt_D} + h_{mD} = h_D \tag{29}$$

The solution of (29) in Laplace space is

$$\bar{h}_{mD} = \frac{\bar{h}_D}{1 + W'_D p} \tag{30}$$

where $\bar{h}_D$ is defined by (26). Numerical inversion of (30) gives the theoretical response for the observation piezometer in terms of dimensionless drawdown.

Theoretical Responses

In this section, theoretical responses, generated using realistic aquifer parameters, are plotted to demonstrate the influence of pumped well characteristics on the determination of

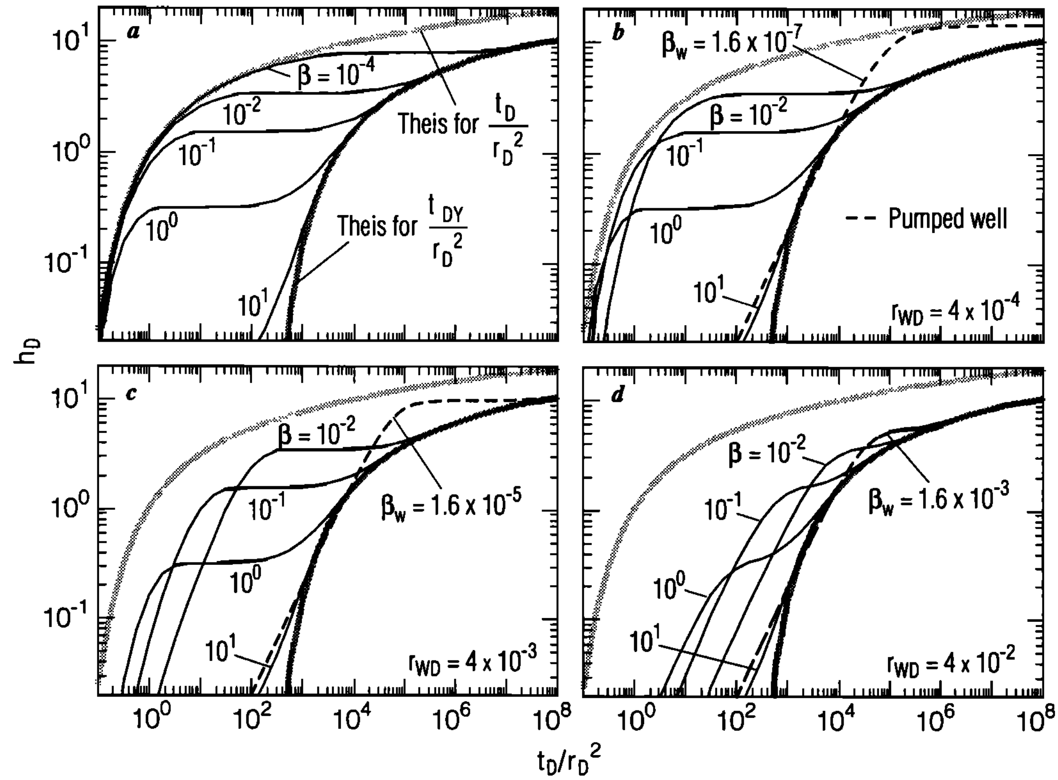

Figure 2. Dimensionless type curves for fully penetrating pumped and observation wells with $\sigma = 2 \times 10^{-4}$ and the indicated values of β showing (a) *Neuman*'s [1975] universal type curves for a well of infinitesimal diameter and, in the remaining figures, type curves for a well of finite diameter with $W_D = 10^4$ in a 25 m thick aquifer with the indicated values of β_w and r_{wD}, where (b) $r_w = 0.01$ m, (c) $r_w = 0.1$ m, and (d) $r_w = 1.0$ m.

specific storage. It is assumed, for simplicity, that no well bore skin is present ($S_w = 0$). It should be noted, however, that effects of well bore skin can be important in the interpretation of pumped well data and early time piezometer responses. The reader is referred to *Moench* [1985, Figures 7 and 8] for illustrations of the effects of well bore skin in the analogous case of flow to a finite-diameter well in a leaky aquifer. It is also assumed, for simplicity, that water is released instantaneously from the unsaturated zone ($\gamma = \infty$).

Fully Penetrating Pumped and Observation Wells

Figure 2a is a set of *Neuman*'s [1975] "universal" type curves (wells are infinitesimal in diameter) for $\sigma = 2 \times 10^{-4}$ and for the indicated values of β. (Theis type curves corresponding to *Neuman*'s [1975] type A (for early time response) and type B (for late time response), are shown in Figures 2 and 3 to serve as a basis for comparison.) Figures 2b, 2c, and 2d illustrate type curves for the same values of σ and β as those in Figure 2a except that the type curve for $\beta = 10^{-4}$ has been replaced by type curves for the β_w, which take on values corresponding to different pumped well diameters in each figure. The dimensionless groupings are obtained by assuming $b = 25$ m, $K_z/K_r = 1$, $S_y = 0.25$, $S_s = 2 \times 10^{-6}$ m^{-1}, and four values of r equal to 2.5, 7.9, 25, and 79 m. It is also assumed that $r_c = r_w$. The pumped well diameter (r_w) is assigned values of 0.01, 0.1, and 1.0 m, respectively, for Figures 2b, 2c, and 2d. The values of the dimensionless parameters are indicated in the figures.

A superposition of any of the curves of Figures 2b, 2c, or 2d on Figure 2a reveals that the late time parts of Figures 2b, 2c, and 2d exactly match the corresponding parts of Figure 2a. It is clear from these illustrations that the early time part of *Neuman*'s [1975] universal type curves (type A curves) does not manifest itself if the pumped well has a finite radius. Even with an unrealistically small pumped well radius of 0.01 m (Figure 2b), the early time responses depart significantly from the Theis curve. It is evident from Figure 2 that as the pumped well radius decreases, the solution for a fully penetrating well approaches the line source solution of *Neuman* [1972]. Features similar to those in Figure 2 are demonstrated by *Narasimhan and Zhu* [1993] using a numerical model.

Figure 3 shows the effect of varying W_D while holding r_{wD} constant, whereas Figure 2 shows the effect of varying the dimensionless well radius (r_{wD}) while holding the dimensionless well bore storage (W_D) constant. In Figure 3a, $r_w = r_c = 0.25$ m, and in Figure 3b, $r_w = 0.25$ m and $r_c = 0.005$ m; consequently, $W_D = 10^4$ in Figure 3a, and $W_D = 4$ in Figure 3b. The remaining parameters in Figure 3 are the same as in Figure 2. Figure 3 illustrates graphically that as the well bore storage parameter (controlled by the ratio r_c/r_w) decreases, the solution approaches the solution of *Neuman* [1972]. Comparison of Figures 3a and 3b reveals differences in the shape of the responses in both the pumped well and observation wells that could be diagnostic of the specific storage parameter if accurate early time data are available.

Hypothetical Case Study

Figure 4 is a cross section of a hypothetical, homogeneous water table aquifer illustrating the position of the well screen

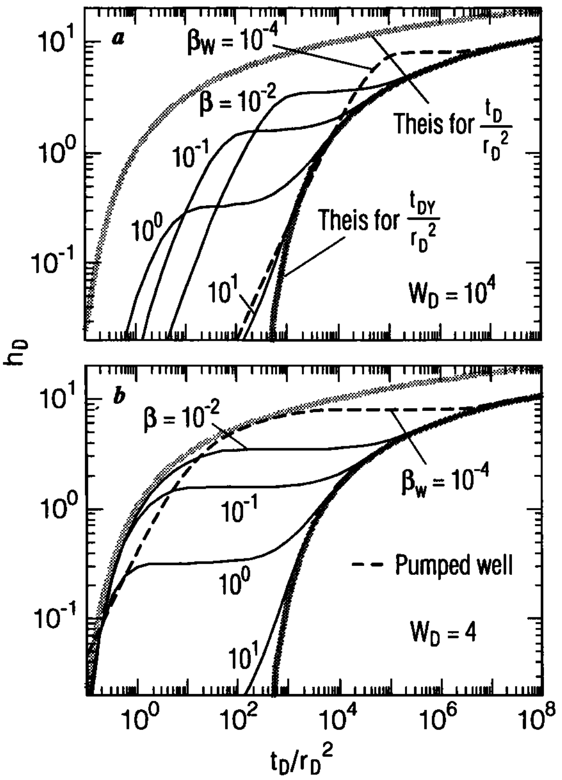

Figure 3. Dimensionless type curves for a fully penetrating pumped well with $r_w = 0.25$ m in a 25 m thick aquifer with $\sigma = 2 \times 10^{-4}$, $r_{wD} = 0.01$, and the indicated values of β, β_w, and W_D where (a) $r_c = 0.25$ m and (b) $r_c = 0.005$ m.

and the locations of four piezometers. It is initially assumed that effects of delayed piezometer response have been effectively eliminated with hydraulic packers. Aquifer parameters and pumped well characteristics are given in Table 2, and piezometer positions are given in Table 3. Water is assumed to be released instantaneously from the unsaturated zone ($\alpha_1 = \infty$), and well bore skin is assumed to be negligible ($S_w = 0$). Parameters are chosen to represent realistic conditions for a clean, well-sorted, homogeneous, slightly anisotropic, medium-grained sand. The chosen value of specific storage is believed by this author to be a realistic estimate for compacted, coarse-grained, unconsolidated, alluvial-aquifer material. This view is supported by field studies involving aquifer-system compaction and tidal response studies [see *Ireland et al.*, 1984, Table 4; *Heywood*, 1995].

Table 2. Parameters for the Hypothetical Example

Parameter	Value
K_z	0.5×10^{-4} m s^{-1}
K_r	1×10^{-4} m s^{-1}
S_y	0.2
S_s	2×10^{-6} m^{-1}
b	10 m
r_w	0.1 m
r_c	0.1 m
Q	2×10^{-3} m^3 s^{-1}
l	10 m
d	5 m
S_w	0
α_1	∞

A finite difference model was used to compute drawdown in the pumped well and in the four ideal (no delayed response) piezometers. By so doing, it is possible to verify the analytical solution and at the same time use the simulated data for evaluation of the aquifer constants. The numerical results were kindly provided by T. Reilly (written communication, 1996), who used the "preprocessor" model of *Reilly and Harbaugh* [1993] to generate input to the USGS Modular Ground-Water Flow Model [*McDonald and Harbaugh*, 1988]. To account for the diameter of the pumped well and effects of well bore storage, Reilly used a very large (effectively infinite) radial conductance between nodes that represent the well at the well screen, zero radial conductance between the nodes representing the well where the well is cased, a large (effectively infinite) vertical conductance inside the pumped well, and a storage capacity for the topmost node in the well (representing the free surface) that corresponds to a unit value of specific yield. The finite difference flow model for the aquifer was designed to obtain an accurate representation of the flow field and to accommodate the positions of the piezometers in Table 3. This

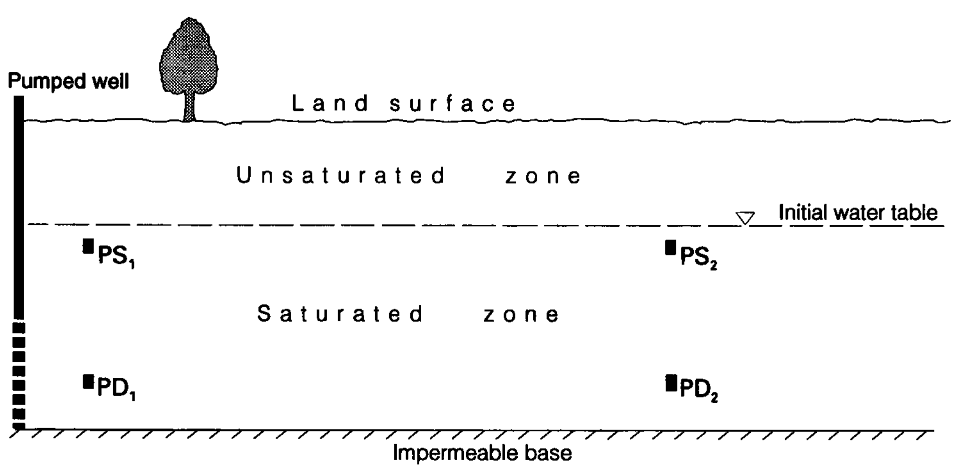

Figure 4. Cross section of the hypothetical aquifer with a partially penetrating pumped well of finite diameter and the locations of four points of observation. The aquifer parameters and well dimensions are indicated in Table 2. (Drawn approximately to scale.)

Table 3. Positions of Piezometers for Hypothetical Example

Piezometer	r, m	r_D	Depth,* m	z_D†
PD_1	3.16	31.6	7.5	0.25
PD_2	31.6	316.	7.5	0.25
PS_1	3.16	31.6	1.	0.90
PS_2	31.6	316.	1.	0.90

*Depth of piezometer opening below initial water table.
†Vertical position measured from base of aquifer.

was accomplished using 30 equally spaced (with two minor exceptions) rows and 80 columns that increase in width with distance from the well by a factor of 1.1103. Computed drawdown is tabulated in Table 4, where results are compared with values obtained by the model proposed in this paper. The greatest differences between the numerical model and the analytical model occur in the piezometer closest to the pumped well screen and in the pumped well itself. Differences can likely be attributed to simplifying assumptions (e.g., uniform well face flux) used in the well bore condition for the analytical model, whereas in the finite difference model, flow across the well screen can vary with vertical position. In addition, there are minor errors due to discretization in the numerical model and computer precision in the evaluation of the analytical solution by numerical inversion. Additional detailed comparison of the analytical and numerical models might prove useful for determining ramifications of the assumptions imbedded in the well bore boundary condition (4a).

Parameter estimation by the Neuman model. Applying the *Neuman* [1974] model and following the methodology described by *Moench* [1994], analysis of the numerical data in Table 4 yields the matches shown in Figures 5a and 5b. The type-curve matching procedure yields a value of $K_D = 0.5$, in agreement with the value in Table 2. Thus $\beta = 0.05$ for the piezometers located near the pumped well, and $\beta = 5.0$ for the more distant piezometers. In this analysis, since the β values are now known, it remains only to determine the optimal value of σ. (Match points are chosen for this and subsequent figures using $t_D/r_D^2 = 1$ and $h_D = 1$ as the match-point coordinate of the dimensionless type curves.) The chosen match point in Figure 5a is $t/r^2 = 2.0\ s/m^2$ and $h - h_t = 0.16$ m, and the chosen match point in Figure 5b is $t/r^2 = 0.20\ s/m^2$ and $h - h_t = 0.15$ m. The difference in the horizontal location of the match point is due to the fact that the match in

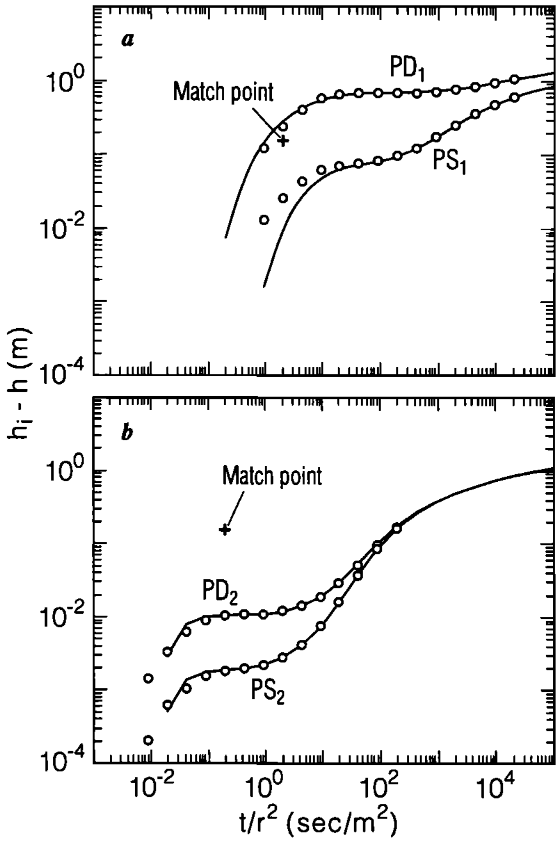

Figure 5. Composite plots of drawdown in the hypothetical aquifer (open circles) fitted with a single match point to *Neuman* [1974] type curves for piezometers located (a) at $r = 3.16$ m and (b) at $r = 31.6$ m.

Table 4. Hypothetical Drawdown Versus Time for the Pumped Well, Two Deep Piezometers, and Two Shallow Piezometers

	Pumped Well		PD_1*		PD_2*		PS_1†		PS_2†	
Time	A	N	A	N	A	N	A	N	A	N
9.28×10^0	0.530	0.516	0.128	0.125	0.0014	0.0014	0.013	0.013	0.0002	0.0002
2.00×10^1	1.02	0.99	0.254	0.244	0.0034	0.0033	0.025	0.026	0.0006	0.0006
4.31×10^1	1.74	1.67	0.437	0.416	0.0063	0.0061	0.044	0.044	0.0011	0.0010
9.28×10^1	2.41	2.33	0.606	0.584	0.0090	0.0088	0.062	0.062	0.0015	0.0015
2.00×10^2	2.70	2.66	0.681	0.665	0.0102	0.0102	0.071	0.072	0.0018	0.0018
4.31×10^2	2.74	2.71	0.693	0.680	0.0106	0.0106	0.075	0.077	0.0019	0.0019
9.28×10^2	2.74	2.71	0.695	0.683	0.0110	0.0110	0.082	0.084	0.0022	0.0022
2.00×10^3	2.75	2.72	0.701	0.690	0.0120	0.0120	0.096	0.098	0.0028	0.0028
4.31×10^3	2.76	2.74	0.715	0.703	0.014	0.014	0.123	0.126	0.0042	0.0042
9.28×10^3	2.79	2.76	0.741	0.729	0.019	0.019	0.174	0.177	0.0074	0.0075
2.00×10^4	2.84	2.81	0.788	0.775	0.029	0.029	0.257	0.258	0.016	0.016
4.31×10^4	2.91	2.89	0.861	0.847	0.052	0.052	0.366	0.366	0.037	0.038
9.28×10^4	3.01	2.98	0.957	0.942	0.100	0.098	0.488	0.487	0.086	0.085
2.00×10^5	3.12	3.09	1.068	1.052	0.175	0.172	0.613	0.612	0.166	0.163

Drawdown is measured in meters; time is measured in seconds; A is analytical results; and N is numerical results.
*Deep piezometers.
†Shallow piezometers.

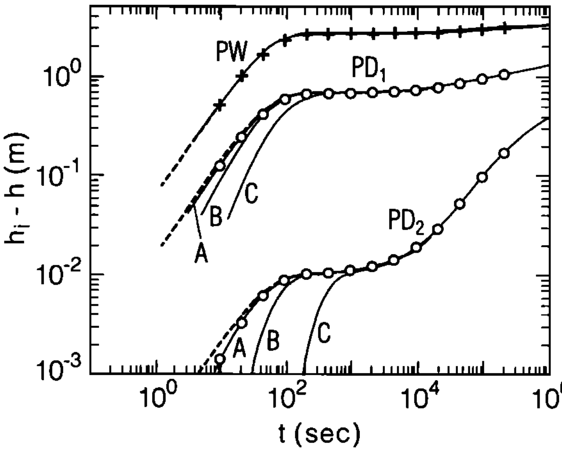

Figure 6. Drawdown versus time in the pumped well (pluses) and at points PD_1 and PD_2 in the hypothetical aquifer (open circles) compared with drawdown obtained with the proposed model for $S_s = 2 \times 10^{-6}$ m^{-1} (A curves), $S_s = 2 \times 10^{-5}$ m^{-1} (B curves), and $S_s = 2 \times 10^{-4}$ m^{-1} (C curves). (Dashed lines show the responses for a rigid aquifer ($S_s = 10^{-10}$ m^{-1}).)

Figure 5a is achieved by using an approximate value of $\sigma = 10^{-2}$ and the match in Figure 5b is achieved by using an approximate value of $\sigma = 10^{-3}$. These values of σ yield values of specific storage of 2×10^{-4} m^{-1} and 2×10^{-5} m^{-1} for the near and distant points, respectively. With the exception of specific storage, all the parameters indicated in Table 2 are accurately reproduced with the Neuman model. The values of S_s, however, are clearly unacceptable, being off by factors of 10 and 100.

Influence of specific storage on drawdown response. The *Neuman* [1974] model can be used to evaluate S_s only if precautions are taken to eliminate effects of well bore storage in the pumped well (and effects of delayed piezometer response). Under the influence of well bore storage the proposed model can, in principle, be used to estimate S_s, but accurate early time data from piezometers located at various distances from the pumped well are needed. Pumped well data are also important for a complete and accurate analysis. Because both W_D and σ depend upon S_s, the usual method of type-curve analysis is found to be rather cumbersome. A preferred approach is to first estimate the parameters K_r, K_z, and S_y by methods of type-curve analysis and then use forward modeling as demonstrated below.

Figure 6 shows the exact match (PW and A curves) of analytical model responses with the synthetic numerical drawdown data using the parameters indicated in Table 2 for the pumped well and the deep-seated points of observation, PD_1 and PD_2. Figure 6 also shows theoretical responses for values of specific storage that are 10 (B curves) and 100 (C curves) times larger than the value given in Table 2, the remaining parameters being unchanged. Drawdown in the pumped well is not noticeably changed by variations in S_s. Reductions in the value of specific storage below 2×10^{-6} m^{-1} bring about no significant change in the match of the theoretical response to the synthetic data (see the dashed lines in Figure 6). This is because hydraulic head at points of observation in the hypothetical aquifer respond almost instantaneously to changing drawdown in the pumped well when S_s is less than about 10^{-6} m^{-1}. Under this condition the response should approach that of a rigid aquifer such as that obtained by *Kipp* [1973]. (Kipp's solution, unfortunately, is not directly comparable as it assumes the aquifer is infinitely thick and includes the potential distribution in the vertical cylinder of aquifer material directly below the base of the well.) Figure 6 shows that increased values of specific storage bring about increased delays in early time responses to pumping. This is due to proportionally decreased hydraulic diffusivity. Thus it should, at least in principle, be possible to estimate this parameter from aquifer-test data by comparing the response in the pumped well with the response in observation piezometers.

In the hypothetical example, well bore skin S_w was taken to be zero to simplify the analysis and discussion. If a skin of low permeability relative to the aquifer material is present, increased drawdown in the pumped well will result. Well bore skin will also bring about increased delay of early time drawdown in the piezometers [see *Moench*, 1985, Figure 7] and will therefore have an effect on estimates of specific storage. A detailed discussion of the influence of well bore skin upon estimates of aquifer parameters is beyond the scope of this paper. It should be noted, however, that the parameter can be determined by trial and error matching of theoretical responses with late time pumped well and observation well data. Once obtained, the determined value of S_w should be included in the model to improve on estimates of the aquifer parameters.

Because of the importance of accurate measurements of early time drawdown in estimating S_s, it is necessary either to account for effects of delayed piezometer response or to eliminate them as much as possible by using a transducer and hydraulic packer [see e.g., *Butler et al.*, 1996]. Figure 6 is illustrative of the latter situation.

Influence of delayed response of piezometers. Figure 7 is similar to Figure 6 but compares the drawdown in the hypothetical aquifer at locations PD_1 and PD_2 (open circles) with the drawdown in piezometers typically used in aquifer tests. Perfect hydraulic connection between the aquifer and the piezometer is assumed. The responses are computed using (26) and (30). The appropriate shape factor [*Hvorslev*, 1951, case 8] is

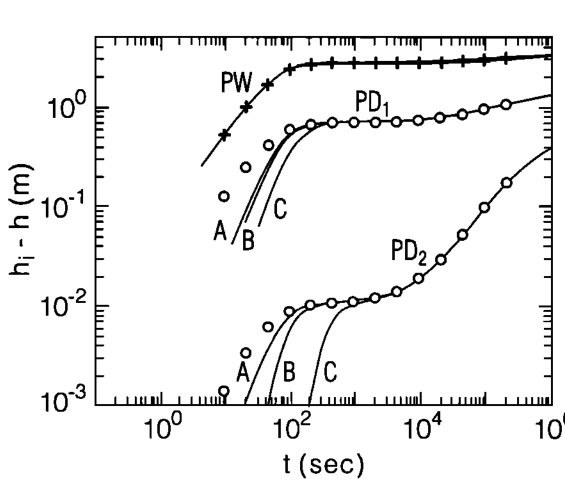

Figure 7. Delayed response of piezometers (solid lines) with $r_p = 0.025$ m and $L = 0.5$ m located at points PD_1 and PD_2 in the hypothetical aquifer for the conditions indicated in Figure 6. Response in the aquifer (open circles) is shown for comparison along with that of the pumped well (pluses); $W_D = 5 \times 10^4$.

$$F' = \frac{L}{\ln\{x + (1 + x^2)^{0.5}\}} \qquad (31)$$

where $x = mL/2r_p$, $m = (K_r/K_z)^{1/2}$, and $L = z_2 - z_1$. The diameter of the piezometers in this hypothetical example is 0.05 m, and the screen length is 0.50 m. The A curves in Figure 7 are computed using the parameters given in Table 2 (where $S_s = 2 \times 10^{-6}\ \mathrm{m}^{-1}$). Figure 7 also shows theoretical responses for values of specific storage that are 10 (B curves) and 100 (C curves) times larger than the value given in Table 2. Careful inspection of Figures 6 and 7 reveals that effects of delayed piezometer response are most important in the piezometer closer to the pumped well (PD_1) where changes in hydraulic head occur most rapidly. At the more distant piezometer (PD_2), curve C in Figure 7, for example, is only slightly displaced to the right in comparison with the corresponding curve in Figure 6. Additional delay in the piezometer responses can result from there being imperfect hydraulic connection between the piezometer and the aquifer such as by screen clogging, aquifer damage brought about by improper piezometer installation, and other factors described in detail by *Hvorslev* [1951]. In practice, it would be best to follow the suggestion of *Black and Kipp* [1977] and check for good hydraulic connection by slug testing the individual piezometers. With an estimate of K_r obtained from the slug test, the value of F' can be estimated directly rather than by means of approximate formulae.

The effects of delayed piezometer response demonstrated in Figure 7 by the A curves are relatively small because of the dominating influence of well bore storage in the pumped well. As one might expect, effects of delayed piezometer response are enhanced relative to the aquifer response when well bore storage in the pumped well is reduced. This is illustrated in Figure 8, which makes a comparison similar to that in Figure 7 (A curves) but with a smaller well bore storage brought about by reducing r_c relative to r_w ($r_c = 0.1r_w$). The well bore storage parameter W_D now becomes 5×10^2 compared with the value of 5×10^4 in Figure 7. Enhanced effects of delayed piezometer response are due to the more rapid increase in aquifer drawdown seen in Figure 8 (open circles).

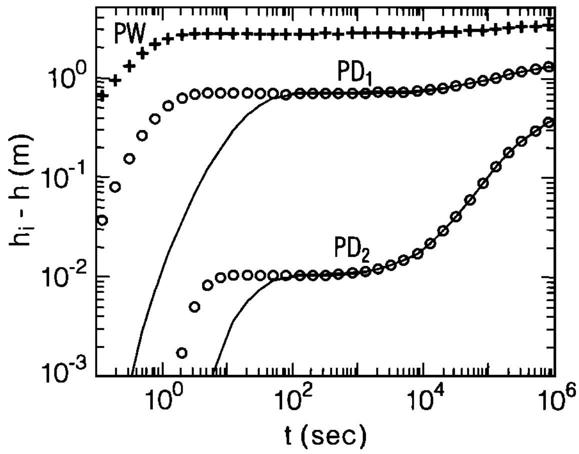

Figure 8. Delayed response of piezometers (solid lines) with $r_p = 0.025$ m and $L = 0.5$ m located at points PD_1 and PD_2 in the hypothetical aquifer for the conditions of the A curves in Figure 7 but with small well bore storage in the pumped well. Response in the aquifer (open circles) is shown for comparison along with that of the pumped well (pluses); $W_D = 5 \times 10^2$.

Summary and Conclusions

The proposed solution for flow to a partially penetrating well of finite diameter in a water table aquifer allows for improved interpretation of early time drawdown data and the possible evaluation of the specific storage parameter. Because of the importance of accurate early time data the solution includes effects of delayed piezometer response. Accurate evaluation of S_s is not possible with the classical line source solutions without eliminating the influence of well bore storage in the pumped well and the effects of delayed piezometer response. Analysis of synthetic data that were generated using a finite difference model of a hypothetical but realistic water table aquifer reveals that estimates of specific storage can be exaggerated by orders of magnitude if a line source model is used to fit early time drawdown data.

The synthetic data from the hypothetical water table aquifer were analyzed with the proposed solution using inverse (type-curve analysis) and forward (sensitivity analysis) methods, and exact values for the hydraulic parameters were obtained. It was shown that if S_s is large, it should be possible to evaluate the parameter by trial and error analysis using forward methods. If, however, the aquifer is not highly compressible (small S_s), the response in observation piezometers located in the vicinity of the pumped well will be nearly simultaneous, and accurate evaluation of S_s will be difficult even with the present model. In this situation, evaluation of the parameter is most readily achieved if steps are taken to reduce effects of well bore storage and delayed piezometer response.

Appendix

This appendix provides a derivation of the Laplace transform solutions, equations (18) and (19). The boundary-value problem in Laplace space is given by (13)–(17).

A solution to (13) that satisfies (16) is

$$\bar{h}_D = \sum_{n=0}^{\infty} \bar{f}_n(r_D, p) \cos[(n + 1/2)\pi z_D] \qquad (A1)$$

where $n = 0, 1, 2, \cdots$. Substitution of (A1) into (13) yields

$$\sum_{n=0}^{\infty} \left\{ \bar{f}_n'' + \frac{1}{r_D}\bar{f}_n' - [(n + 1/2)^2\pi^2\beta_w + p]\bar{f}_n \right\}$$

$$\cdot \cos[(n + 1/2)\pi z_D] = 0 \qquad (A2)$$

hence

$$\bar{f}_n'' + \frac{1}{r_D}\bar{f}_n' - [(n + 1/2)^2\pi^2\beta_w + p]\bar{f}_n = 0 \qquad (A3)$$

the solution of which can be written as

$$\bar{f}_n = u_n(p)K_0(q_n r_D) + v_n(p)I_0(q_n r_D) \qquad (A4)$$

where $q_n = [(n + 1/2)^2\pi^2\beta_w + p]^{1/2}$. K_0 and I_0 are the zero-order modified Bessel functions of the second and first kind, respectively, and u_n and v_n are coefficients to be determined.

Because of (14), $v_n(p) = 0$, and consequently,

$$\bar{f}_n = u_n(p)K_0(q_n r_D) \qquad (A5)$$

Substitution of (A5) into (A1) yields

$$\bar{h}_D = \sum_{n=0}^{\infty} u_n(p) K_0(q_n r_D) \cos[(n + 1/2)\pi z_D] \quad \text{(A6)}$$

The coefficients $u_n(p)$ are determined from (15a) and (15b) by first substituting (A6) into (15a) and letting $r_D = 1$. Thus

$$-\sum_{n=0}^{\infty} u_n(p) q_n K_1(q_n) \cos[(n + 1/2)\pi z_D] = \bar{q}_s \quad \text{(A7)}$$

$$1 - l_D \le z_D \le 1 - d_D$$

where

$$\bar{q}_s = -\frac{2}{p(l_D - d_D)} + W_D p \bar{h}_{wD} \quad \text{(A8)}$$

Likewise, applying (15b)

$$-\sum_{n=0}^{\infty} u_n(p) q_n K_1(q_n) \cos[(n + 1/2)\pi z_D] = 0 \quad \text{(A9)}$$

$$z_D < 1 - l_D; \quad z_D > 1 - d_D$$

Multiplying (A7) through by $\cos[(m + 1/2)\pi z_D]$, where m is an integer, and integrating over the indicated interval, one obtains

$$-\sum_{n=0}^{\infty} u_n(p) q_n K_1(q_n) \int_{1-l_D}^{1-d_D} \cos[(n + 1/2)\pi z_D] \cdot \cos[(m + 1/2)\pi z_D]\, dz_D$$

$$= \bar{q}_s \int_{1-l_D}^{1-d_D} \cos[(m + 1/2)\pi z_D]\, dz_D \quad \text{(A10)}$$

Multiplying (A9) by $\cos[(m + 1/2)\pi z_D]$ and integrating over the interval below the screen, one obtains

$$-\sum_{n=0}^{\infty} u_n(p) q_n K_1(q_n) \int_{0}^{1-l_D} \cos[(n + 1/2)\pi z_D] \cdot \cos[(m + 1/2)\pi z_D]\, dz_D = 0 \quad \text{(A11)}$$

Also, by performing the same operation over the interval above the screen, one obtains

$$-\sum_{n=0}^{\infty} u_n(p) q_n K_1(q_n) \int_{1-d_D}^{1} \cos[(n + 1/2)\pi z_D] \cdot \cos[(m + 1/2)\pi z_D]\, dz_D = 0 \quad \text{(A12)}$$

Adding (A11) and (A12) to the left-hand side of (A10), one obtains

$$-\sum_{n=0}^{\infty} u_n(p) q_n K_1(q_n) \int_{0}^{1} \cos[(n + 1/2)\pi z_D] \cdot \cos[(m + 1/2)\pi z_D]\, dz_D$$

$$= \bar{q}_s \int_{1-l_D}^{1-d_D} \cos[(m + 1/2)\pi z_D]\, dz_D \quad \text{(A13)}$$

The set $\{\cos[(n + 1/2)\pi z_D]\}$ can be shown by direct integration to be orthogonal over the interval 0, 1. Thus all terms in the sum on the left-hand side of (A13) are zero except those for which $n = m$. (For a discussion on the topic of orthogonality, see *Hildebrand* [1976, p. 207] or *Churchill* [1941, p. 38].) As a consequence of this property, (A13) becomes

$$-u_n(p) q_n K_1(q_n) \int_0^1 \cos^2[(n + 1/2)\pi z_D]\, dz_D$$

$$= \bar{q}_s \int_{1-l_D}^{1-d_D} \cos[(n + 1/2)\pi z_D]\, dz_D \quad \text{(A14)}$$

Performing the integration and rearranging terms, one obtains

$$u_n(p) = -2\bar{q}_s\{\sin[(n + 1/2)\pi(1 - d_D)] - \sin[(n + 1/2)\pi(1 - l_D)]\} \cdot \{q_n K_1(q_n)[(n + 1/2)\pi + 0.5 \sin(2(n + 1/2)\pi)]\}^{-1} \quad \text{(A15)}$$

Now, applying the solution (A6) to (17) and evaluating the result at $z_D = 1$, one obtains

$$-[(n + 1/2)\pi] \tan[(n + 1/2)\pi] = -\frac{p}{\sigma\beta + p/\gamma} \quad \text{(A16)}$$

Thus the solution (A6) becomes

$$\bar{h}_D = \sum_{n=0}^{\infty} u_n(p) K_0(q_n r_D) \cos[\varepsilon_n z_D] \quad \text{(A17)}$$

where

$$u_n(p) = -\frac{2\bar{q}_s \sin[\varepsilon_n(1 - d_D)] - \sin[\varepsilon_n(1 - l_D)]}{q_n K_1(q_n)[\varepsilon_n + 0.5 \sin(2\varepsilon_n)]} \quad \text{(A18)}$$

and ε_n are the roots of $\varepsilon_n \tan(\varepsilon_n) = p/(\sigma\beta_w + p/\gamma)$ for $n = 0, 1, 2, \cdots$. Thus

$$\bar{h}_D(r_D, z_D, p) = -\bar{q}_s E \quad \text{(A19)}$$

where E is defined by (21).

The hydraulic head in the pumped well is obtained by inserting (A19) into (15d). Thus, at $r_D = 1$, one obtains

$$\bar{h}^*_D = -\bar{q}_s A \quad \text{(A20)}$$

where A is defined by (20). From (15c) and (A20),

$$\bar{h}_{wD} = -\bar{q}_s A - S_w \bar{q}_s \quad \text{(A21)}$$

Substitution of (A8) into (A21) yields the solution (18) for the dimensionless head in the pumped well.

Substituting (A21) into (A8) and combining the result with (A19) yields the solution (19) for the dimensionless head in the aquifer.

Notation

- b initial saturated thickness of aquifer, L.
- d vertical distance from initial water table to top of pumped well screen, L.

d_s thickness of the well bore skin, L.
F' modified Hvorslev shape factor, L.
h hydraulic head, L.
h^* vertical average of hydraulic head in the aquifer adjacent to and over the length of the pumped well screen, L.
h_i initial hydraulic head, L.
h_m measured hydraulic head in a piezometer, L.
h_w measured hydraulic head in the pumped well, L.
K_z hydraulic conductivity in the vertical direction, $L\ T^{-1}$.
K_r hydraulic conductivity in the horizontal direction, $L\ T^{-1}$.
K_s hydraulic conductivity of the well bore skin, $L\ T^{-1}$.
l vertical distance from initial water table to bottom of pumped well screen, L.
L length of a piezometer screen, L.
Q pumping rate, $L^3\ T^{-1}$.
p Laplace transform variable.
r radial distance from axis of pumping well, L.
r_w outside radius of the pumped well screen, L.
r_c inside radius of the pumped well over the range of changing water levels during pumpage, L.
r_p inside radius of an observation piezometer over the range of changing water levels during pumpage, L.
S storativity.
S_y specific yield.
S_s specific storage, L^{-1}.
T transmissivity, $L^2\ T^{-1}$.
t time since start of pumping, T.
z vertical distance above bottom of aquifer, L.
α_1 empirical constant for drainage from the unsaturated zone, T^{-1}.

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